

❑ 15 Descriptive Statistics - Three variables

Most of the time when we are exploring patterns or associations in data it will be sufficient to consider two variables at a time. Occasionally, we might want to look at three variables at once. One of the reasons why we might want to do so is to help us identify if a confounding variable might be in play.

To explain what confounding is we consider again the titanic data. We have already seen using charts that there is an association between sex and survival (a greater percentage of women compared to men survived). We have also seen an association between class and survival (the survival rate went down as we went from 1st to 2nd to 3rd class). But perhaps there is really only one pattern and not two to be observed here. Suppose a very large proportion of women were travelling first class, then the reason why such a high proportion of women survived may be because they were travelling first class and not because they were women?

So was class a factor in survival? Or is it sex that is a factor in survival? Or do both have an effect? We can help address this question with a plot for three categorical variables.

• Charts for three variables – all categorical

Multiple stacked bar chart

The chart opposite (SPSS) shows two clustered bar chart with class defining the clusters and within each cluster, the number that survived are represented by distinct bars (red = did survive, blue = did not survive)

The top clustered bar chart represents females, the bottom represents males.

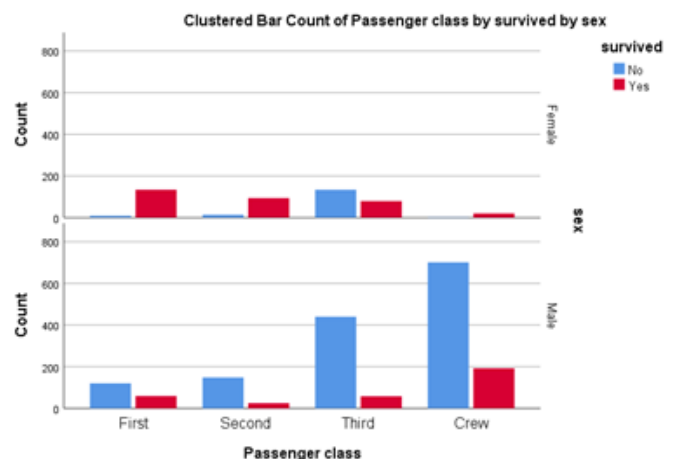
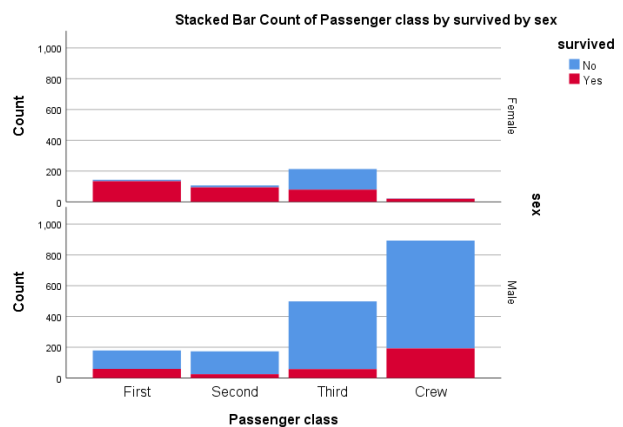
By examining this chart carefully we can see that both class and (rather than just one of these factors) have an influence on survival.

Why can we say that class affects survival? Because if we consider only females then the ratio of red to blue (i.e., survived to did not survive) diminished from most red for 1st and 2nd class passengers to relatively little red for third class passengers. And if we do a similar analysis, now looking only at males, we see a similar pattern.

Why can we say that sex affects survival? Because if we compared the amount of red in the bar representing females (top chart) there is much more than in the bar representing males (bottom chart). And similarly for other classes.

Multiple clustered bar chart

We can establish the same finding using a multiple, clustered bar chart (opposite – SPSS) but it is a little more difficult to do so.

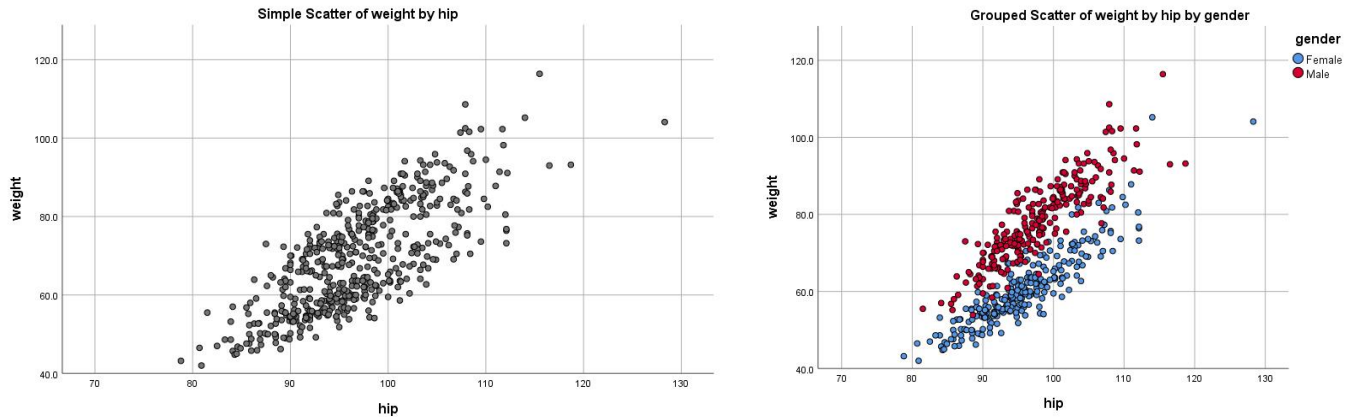


• Charts for three variables – one categorical, two scale

Clustered scatterplot

Example

Suppose we have data for a group of adults recoding both their weights and a measure of their hip circumference. We could show the relationship between these two variables using a scatterplot (below left – SPSS). But it may be more informative to see the relationship between these two variables separately for males and females. We can do this with a clustered scatterplot (below right – SPSS)



We can tell from either plot that weight tends to increase with hip measure. But we extract a couple of bits of additional information from the second plot that we cannot see in the first.

- For a given hip size, men tend to be heavier.
- Men's weights will tend to increase more than women per unit change in hip size measure

Charts for three variables – two categorical, one scale

We consider another problem here we will try to use a trio of charts to address a potential confounding problem

Example

The HR manager of a manufacturing firm is examining employee salaries. For each employee, in addition to their salary he notes their gender (male, female), their experience (years working in the firm) and their role in the firm (operator, technician or engineer).

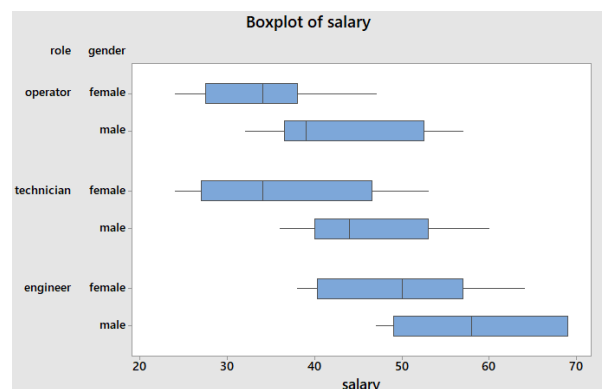
It is to be expected that salary might vary by role (engineers tend to earn more than technicians who in turn, tend to make more than operators). But it also observed that males tend to be paid more than females.

One explanation for this is that there may be gender-based discrimination at play in the firm. An alternative explanation is that there may be a disproportionate number of women in the roles of technician and operator and their salaries are lower as a result. We can address this using a multiple clustered boxplot.

Clustered boxplot (two categorical variables)

The plot opposite is a clustered boxplot with a cluster of two bars (one for males, one for females) at each of the three levels of role (operator, technician, engineer).

We can see from the plot that men do tend to be paid more for each of the three roles being considered. Thus, the theory that gender based discrimination is at play does appear to be a more plausible explanation.



Charts for three variables – all scale

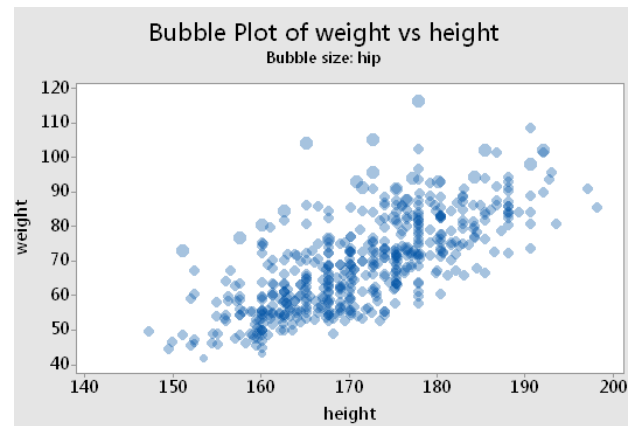
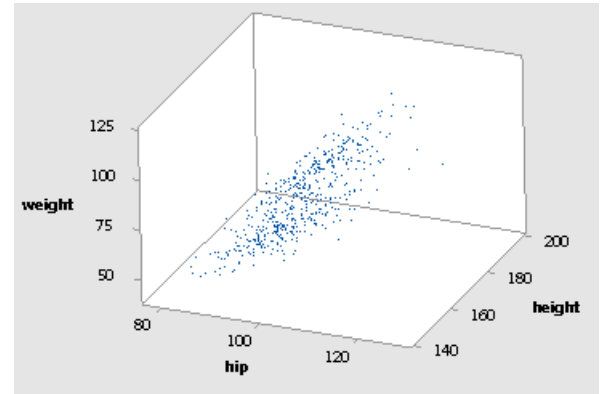
An obvious chart to consider is the 3D scatter plot. Such a plot of weight, hip and height for a sample of people is shown opposite (Minitab).

The points do form a linear pattern in 3D space but it is difficult to determine the exact direction of this line.

Arguably a better option might be to take projections of this plot. I.e., construct 2D scatter plots of the three variables.

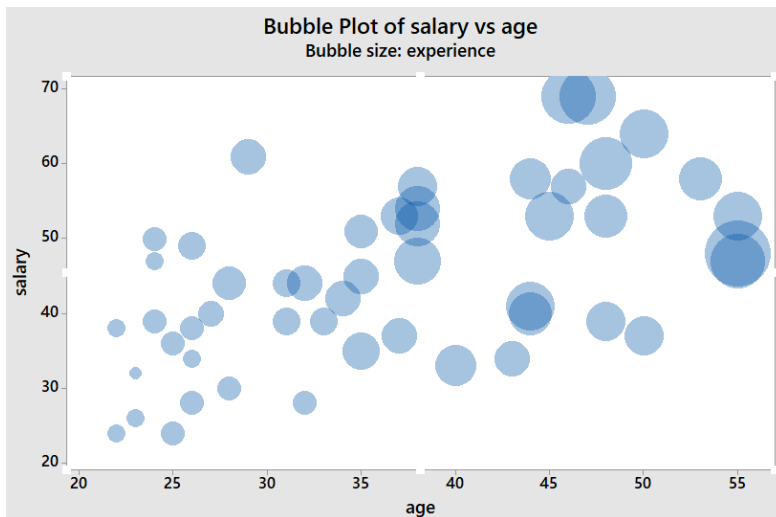
The bubble plot is another way to visualise three scale variables. Weight and height are plotted opposite (Minitab) as in a 2D scatter plot. But each subject is represented by a bubble with an area proportional to that subject's hip size.

Note that smaller bubbles tend to be left and down, larger ones tend to be right and up. This illustrates the same pattern as the 3D scatter plot.



Example

A bubble plot of salary v age with bubble representing salary is shown for the manufacturing firm's salary data.



We can see some tendency for salary to be correlated with age (as we would probably expect) as the points (bubbles) tend to drift from bottom left to top right.

We also see that older employees tend to be more experienced as the bubbles tend to get bigger as you move from left to right on the chart.

And we see that higher paid employees tend to be more experienced as the bubbles tend to get bigger as you move from bottom to top on the chart.